
Quantifying the Effect of Test Set Contamination on Generative Evaluations

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Abstract

Test set contamination – the inclusion of benchmarks in pretraining data – is a critical threat to the trustworthy evaluation of frontier AI systems. While its impact on *discriminative* evaluations is well-studied, contamination on *generative* evaluations remains underexplored. We quantitatively assess these effects across the language model lifecycle by pretraining models (up to 344M parameters) on web data contaminated with varying numbers MATH test set replicas. Performance improves with contamination and model size. Using scaling laws, we make a surprising discovery: including even a single test set replica enables models to achieve lower loss than the irreducible error of training on the uncontaminated corpus. We then study additional training: overtraining with fresh data reduces the effects of contamination, whereas supervised finetuning on the training set will increase performance for low contamination but decrease performance for high contamination. Finally, at inference, we identify factors that modulate memorization: high sampling temperatures mitigate contamination effects, and longer solutions are exponentially more difficult to memorize than shorter ones, presenting a contrast with discriminative evaluations, where solutions are only a few tokens in length. By characterizing how generation and memorization interact, we highlight new considerations for trustworthy evaluation of AI systems.

1. Introduction

As frontier AI systems are pretrained on web-scale data, test set contamination has become a critical concern for accurately assessing their capabilities (Sainz et al., 2023;

Schaeffer, 2023; Xu et al., 2024a; Deng et al., 2024a; Reuel et al., 2025). Evaluation aims to measure generalization on tasks the model has never seen, yet the sheer scale of modern pretraining makes such contamination increasingly likely (Brown et al., 2020; Du et al., 2022; Wei et al., 2022a; Chowdhery et al., 2022; Touvron et al., 2023; Penedo et al., 2024).

Prior research has sought to quantify the impact of test set contamination, also known as leakage, through two primary lenses. *Statistical approaches* attempt to detect contamination or estimate its influence by modifying the test set, for example, by reordering, rephrasing, or replicating benchmark problems, e.g., (Oren et al., 2023; Ni et al., 2025; Shi et al., 2024; Golchin & Surdeanu, 2023; 2024; Roberts et al., 2024; Wang et al., 2025; Zhang et al., 2024a). In comparison, *controlled approaches* – which offer the most rigorous measurement – intentionally contaminate pretraining corpora to quantify how specific doses of leakage inflate performance, e.g., (Magar & Schwartz, 2022; Jiang et al., 2024; Oren et al., 2023; Yao et al., 2024; Duan et al., 2024; Wang et al., 2025; Kocyigit et al., 2025; Bordt et al., 2025; Wei et al., 2025). For a more in-depth discussion, please see Appendix A Related Work.

While foundational, these investigations have predominantly focused on *discriminative* benchmarks like classification or multiple-choice question-answering (MCQA). For example, Magar & Schwartz (2022) used SST-2 (Socher et al., 2013) (classification). Jiang et al. (2024) used SST-2 (classification), MMLU (Hendrycks et al., 2021a) (MCQA), SQuAD (Rajpurkar et al., 2016) (MCQA), and CNN/Daily Mail (fill-in-the-middle) (Nallapati et al., 2016). Oren et al. (2023) used 7 MCQA benchmarks and 1 mathematical problem solving benchmark (GSM8K) (Cobbe et al., 2021). Yao et al. (2024) used 3 MCQA benchmarks while Bordt et al. (2025) used 7 MCQA benchmarks.

However, with the rapid advancement of model capabilities and the advent of reasoning models (OpenAI et al., 2024a; Google Gemini Team et al., 2025; Xu et al., 2025), the field is shifting towards benchmarks for which the model must generate an answer rather than choose between provided answers. Whether test set contamination has the same effect on generative evaluation as on discriminative evaluations

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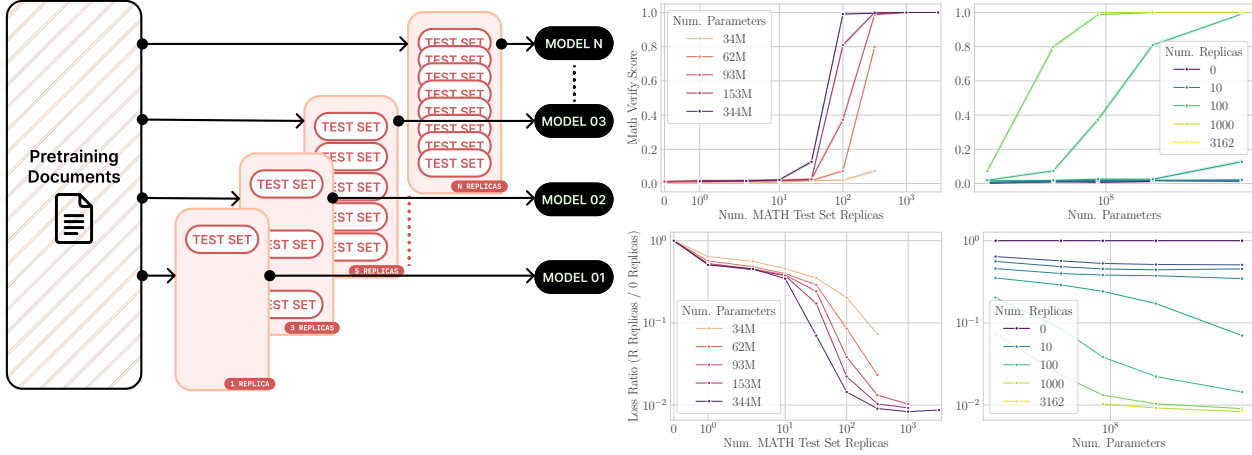


Figure 1. Performance on Generative Benchmarks Increases with Test Set Contamination and Model Size. Schematic: We pretrained compute-optimal language models (34M–344M parameters) on corpora containing different replicas of the MATH test set (0–3162). Evaluation used greedy decoding (temperature = 0). **Left:** As contamination (quantified by the number of test set replicas in the pretraining corpus) increases, Math Verify scores (top) rise and cross entropies on the test set (bottom) fall, consistent with discriminative evaluations, with a sharp improvement around 100 test set replicas. **Right:** The ratio in the loss on the MATH test set between R replicas and 0 replicas grows with model size, meaning larger models benefit more from test set contamination for the same number of replicas (consistent with results that larger models memorize more readily; Tirumala et al., 2022; Carlini et al., 2023; Biderman et al., 2023).

is unclear. Discriminative evaluations typically require the model to place higher probability mass on the correct choice than on a small number of alternative incorrect choices (Gao et al., 2024; Schaeffer et al., 2025d), and candidate choices are often only a couple of tokens long. In contrast, generative evaluations require the model to produce solutions spanning tens-to-thousands of tokens without straying from the memorized path, and introduce new considerations such as the sampling temperature (Ackley et al., 1985), the sampling algorithm (e.g., top-k (Fan et al., 2018), top-p (Holtzman et al., 2020)) and the solution length.

In this work, we quantitatively study the effects of test set contamination on generative evaluations, focusing on the widely used MATH benchmark (Hendrycks et al., 2021b). We pretrain dozens of language models on corpora contaminated with varying numbers of test set replicas, sweeping across model sizes, sampling temperatures, and token budgets. We make the following contributions:

- **Pretraining (Scaling & Irreducible Error):** We quantify how contamination impacts pretrained models. We find that performance increases with the number of test set replicas in the pretraining corpus and with model size, similar to discriminative settings. Under standard scaling law assumptions, we discover that including even a single replica of the test set enables models to achieve a lower loss than the estimated irreducible error of the uncontaminated corpus.
- **More Training (Overtraining & Supervised Finetuning):** We find that training beyond compute-optimal with fresh data dilutes increased performance

from contamination, similar to discriminative settings. We then show that SFT on the training data has opposing effects, depending on the amount of contamination in pretraining: performance improves for low contamination, but worsens for high contamination.

- **Inference (Temperature & Solution Length):** We identify distinct factors that modulate memorization during generation: increasing sampling temperature and increasing solution sequence length each reduce model performance, as models struggle to regurgitate long sequences without decohering. We mathematically describe two competing forces: an exponentially decaying probability to stay on the path of the memorized solution vs an exponentially increasing probability of producing the correct next token.
- **Correction to Evaluation Library:** We identify and fix a critical implementation error in the widely used EleutherAI LM Evaluation Harness (Gao et al., 2024) for Math Verify Scores (Appendix C). Our correction ensures that valid reference solutions are accurately scored as correct (raising gold reference solutions’ scores from $\sim 70\%$ to 100%), a necessary step for trustworthy reporting on mathematical benchmarks.

Our investigation follows the standard language model lifecycle: we first examine contamination during pretraining (Sec. 3), then study how subsequent training—both overtraining and supervised finetuning—interacts with contamination (Sec. 4), and finally analyze how inference-time parameters like sampling temperature and output length modulate performance during deployment (Sec. 5).

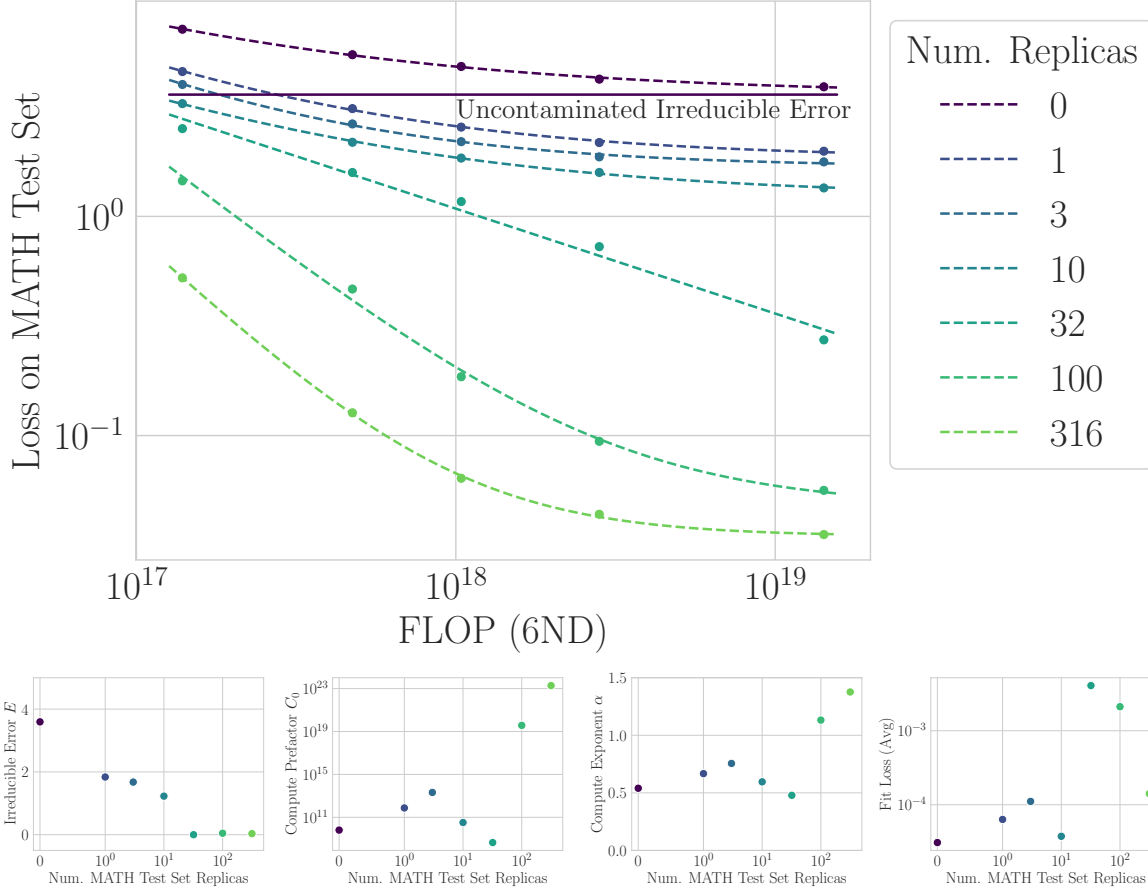


Figure 2. Scaling Laws Suggest Including A Single Test Set Replica Achieves Lower Loss Than the Irreducible Error of the Uncontaminated Corpus. **Top:** For each scaling series pretrained on corpora contaminated with R replicas of the MATH test set, we fit scaling laws $\mathcal{L}(C, R) = E(R) + C_0(R) \cdot C^{-\alpha(R)}$, where $C = 6ND$ is the pretraining compute. Almost all contaminated models achieve lower cross entropy on the MATH test set than the irreducible error of training on uncontaminated data (horizontal purple line). **Bottom:** Increasing test set contamination reduces the irreducible error $E(R)$ from 3.594 at $R = 0$ to 0.0347 at $R = 316$. Larger values of R also increase the compute prefactor and compute exponent. The functional form achieves average fitting error $< 10^{-2}$ for all R .

2. Methods

Pretraining To study the effects of contamination in generative evaluations, we pretrained transformer-based (Vaswani et al., 2017) causal language models from initialization using the Qwen 3 architecture (Yang et al., 2025a), sweeping model sizes: 34M, 62M, 93M, 153M, 344M. Each model was pretrained with 20 tokens-per-parameter (Hoffmann et al., 2022). For each model size and token budget, we created multiple pretraining corpora by contaminating a high quality web crawl corpus (Penedo et al., 2024) with a different number of replicas of the benchmark test set: from 0 (uncontaminated) through 1, 3, 10, 32, 100, 316, 1000, 3162 (logarithmically uniform). Pretraining compute was calculated using the common approximation $C \approx 6ND$ (Kaplan et al., 2020; Sardana et al., 2024; Porian et al., 2024; Gadre et al., 2024), where N is model parameters and D is pre-training tokens. We trained one model for each combination of model size, number of test set replicas, and overtraining

multiplier. For implementation details, see Appendix B.

Benchmark We chose the ubiquitous MATH (Hendrycks et al., 2021b) benchmark of competition math problems due to several properties: it is comparatively large (5,000 test problems), answers exist, and the benchmark includes solutions as well as answers. The MATH test set contains ~ 1.4 M tokens under the Qwen 3 tokenizer.

Evaluation We evaluated our models using two metrics. The first metric we report is *Math Verify* (?), defined as the fraction of problems for which the model generates solutions that are verified to be mathematically equivalent to the benchmark’s boxed answers. We initially evaluated our models using EleutherAI’s Language Model Evaluation Harness (Gao et al., 2024), but discovered a bug in how Math Verify scores are computed by the Harness; for example, the benchmark’s gold reference solutions obtained a Math Verify score of only $\sim 70\%$. We worked with the developers to cor-

rect the implementation (see Appendix C for details). This suggests that research reporting `math_minerva` scores with the Eval Harness prior to task version 3.0 may have reported incorrect scores. The second metric we report is the *Cross Entropy* of the gold reference solutions given the problems, which were previously demonstrated to be useful for studying scaling properties of generative evaluations during pretraining (Schaeffer et al., 2025a). We used temperature-only sampling (?), beginning with temperature 0 (“greedy”), and expanding to more temperatures in Sec. 5.

3. Pretraining: Scaling & Irreducible Error

We begin at the first stage of the model lifecycle: pretraining. We establish foundational results on how test set contamination interacts with model scale and compute, and use scaling laws to quantify what contamination buys a model.

Finding #1: Performance Increases with Contamination and Model Size Consistent with discriminative evaluations, increasing the number of benchmark replicas in the pretraining corpus increases Math Verify scores and decreases cross entropies (Fig. 1 Left), as does increasing model size (Fig. 1 Right). We observe a non-linear relationship between the number of test set replicas and model performance: For low levels of contamination (≤ 10 replicas), the impact on performance is minimal, with Math Verify scores and cross entropies remaining close to uncontaminated performance; at around 100 replicas, performance sharply increases (Fig. 5). At the highest levels of contamination, the model achieves ceiling performance.

Finding #2: Contamination-Driven Performance Is Not Generalization To determine if the performance gains stemmed from robust generalization or superficial memorization, we evaluated our contaminated models on two modified versions of the MATH test set. First, we *rephrased* the problems, retaining the original numerical values and logic but altering the linguistic surface form. Second, we *perturbed* the problems, modifying the numerical values and correct answers while maintaining the problem structure.

In both conditions, performance regresses to match the uncontaminated model across all model sizes and contamination levels in both conditions (Fig. 1). Even models that achieved ceiling performance on the original test set failed to solve the modified versions. This strongly suggests that the capabilities gained from test set contamination rely almost exclusively on verbatim memorization of the specific test sequences rather than the acquisition of underlying mathematical reasoning.

Finding #3: Scaling Laws Suggest Including A Single Test Set Replica Achieves Lower Loss Than the Irre-

Model	Replicas	Rephrased	Perturbed
344M	0	0.04%	0.00%
	1	0.00%	0.00%
	3	0.00%	0.00%
	10	0.00%	0.00%
	32	0.00%	0.00%
	100	0.02%	0.00%
	316	0.00%	0.00%
	1000	0.00%	0.00%
	3162	0.04%	0.00%

Table 1. Contamination-Driven Performance Is Not Generalization. When MATH test problems are rephrased (same numbers, different wording) or perturbed (same wording, different numbers), model performance collapses to baseline regardless of contamination level or model size, confirming that performance gains from contamination are not generalized mathematical reasoning. Results were consistent across all model sizes.

ducible Error of the Uncontaminated Corpus How much does test set contamination “buy” the model in terms of performance? More specifically, how much pretraining compute must be spent on an uncontaminated pretraining corpus to match the performance of a model trained on a corpus containing R replicas of the benchmark test set?

To answer this question, we turned to neural scaling laws. Based on previous work (Kaplan et al., 2020; Hoffmann et al., 2022; OpenAI et al., 2024b; Hu et al., 2024; Schaeffer et al., 2025a), for each scaling series pretrained on corpora contaminated with $R \in \{0, 1, 3, 10, 32, 100, 316\}$ replicas of the MATH test set, we fit neural scaling laws for the cross entropy loss $\mathcal{L}$ on the benchmark test set:

$$\mathcal{L}(C, R) = E_0(R) + C_0(R) \cdot C^{-\alpha(R)}. \quad (1)$$

Fig. 2 shows each scaling law, as well as each scaling law’s estimated parameters as a function of the number of test set replicas R . Our models’ pretraining compute budgets and losses on the test set are reasonably well fit by Eqn. 1, and the pretraining prefactors and pretraining exponents are roughly constant for the various values of R . The biggest effect of increasing contamination is that the irreducible error shrinks from $E(R = 0) = 3.594$ to $E(R = 316) = 0.0347$. On the models we trained, we found that *including even a single replica enables almost all models to achieve lower cross entropy losses than the estimated irreducible error of the uncontaminated pretraining corpus*. Under standard scaling law assumptions – specifically, that the power-law form (Eqn. 1) extrapolates to infinite compute – a contaminated pretraining corpus can buy more than an “infinite” amount of pretraining compute relative to pretraining on an uncontaminated pretraining corpus.

This conclusion potentially contradicts Huang et al. (2024)’s

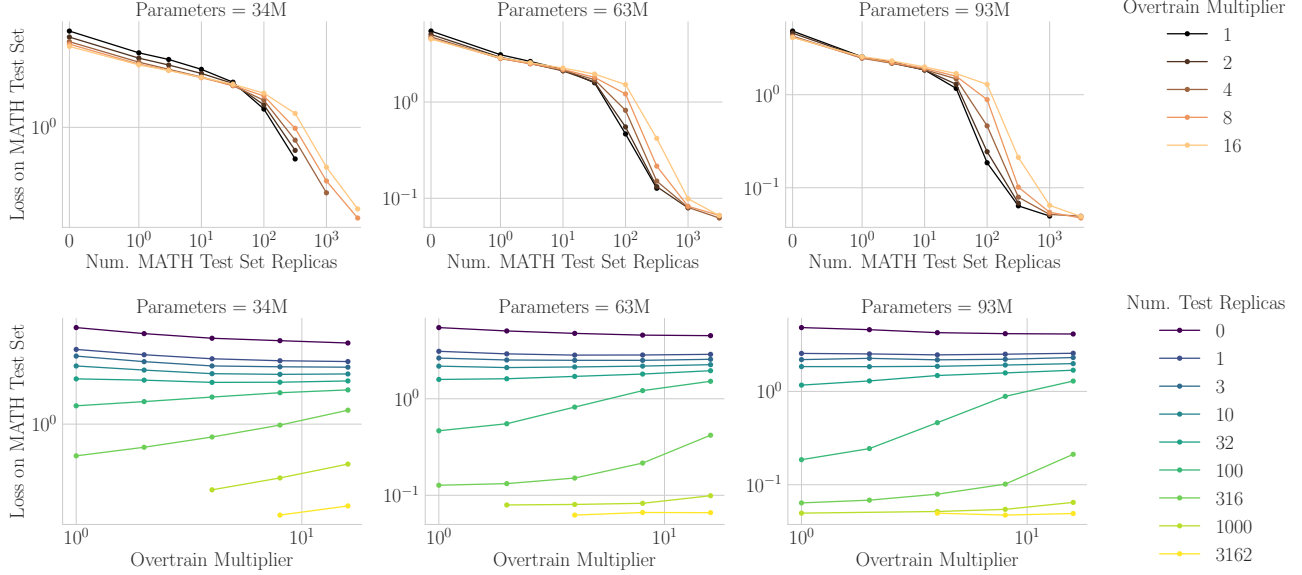


Figure 3. Overtraining with Fresh Data Mitigates Contamination. Consistent with discriminative evaluations (Bordt et al., 2025), we find an interaction between contamination and overtraining (i.e., training longer than Chinchilla compute-optimal; Eqn. 2) on *new fresh data*. For models with low contamination, cross entropy on the MATH test set decreases with increasing overtraining; however, for models with high contamination, cross entropy increases with overtraining. This suggests that while fresh data generally improves performance, it dilutes the “dose,” or proportion of contaminated pretraining tokens, thereby lessening the effect of the contamination. The crossover point shifts with model size, falling from 32 test set replicas for 34M to 1 replica for 93M models, indicating larger models lose their contamination advantage more readily when overtrained.

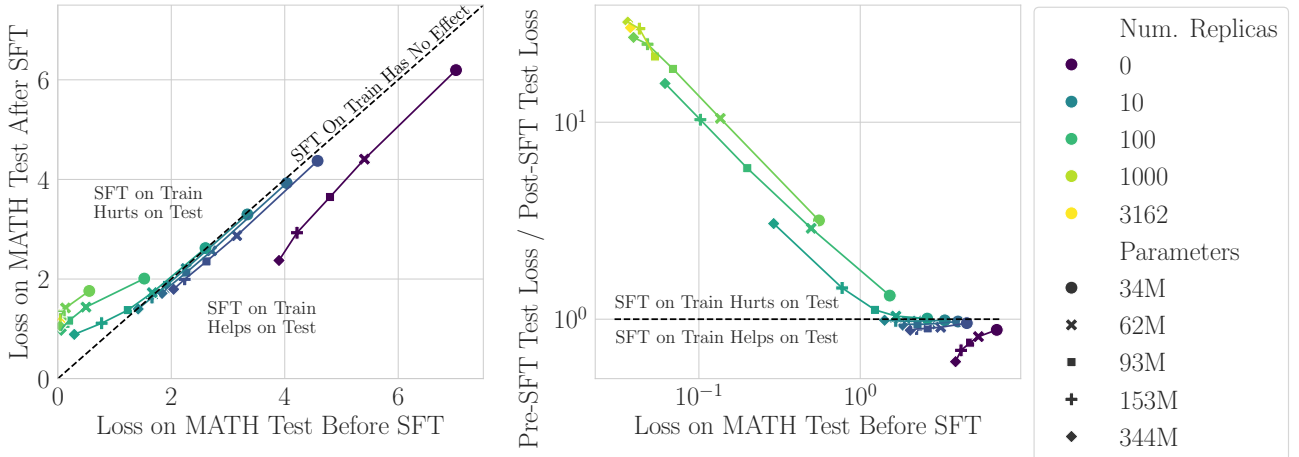


Figure 4. Supervised Finetuning on the Train Set Has Opposing Effects, Depending on Pretraining Contamination. For models pretrained with little-to-no contamination (< 10 test set replicas), supervised finetuning (SFT) on the MATH *train* set decreases loss on the *test* set. For models pretrained with more contamination (> 10 test set replicas), SFT on the train set increases loss on the test set. We conjecture that during SFT, contaminated models learn to generalize but also forget their contaminated pretraining data, and the effects of contaminated test set data are more impactful than generalization for small models, leading to a net increase in test loss.

claim that single-shot verbatim memorization is an “illusion” and Hayes et al. (2025)’s claim that membership inference attacks are limited on pre-trained LLMs, with AUC asymptoting to ~ 0.689 . Future work should aim to understand this difference; one possible explanation is that MATH is distributionally different from FineWeb-Edu-Dedup in a way that

makes identifying test set contamination easier.

4. Further Training: Overtraining & Supervised Finetuning

After pretraining, models typically undergo additional training. We now examine how two common additional training interventions—overtraining on fresh data and supervised finetuning—interact with pretraining contamination.

Finding #4: Overtraining with Fresh Data Mitigates Contamination Bordt et al. (2025) recently reported that for discriminative evaluations, the effect of contamination diminishes when models are trained beyond “compute optimal” on fresh data. We tested whether this so-called *overtraining* (Touvron et al., 2023; Sardana et al., 2024; Gadre et al., 2024; Schaeffer et al., 2025a) has similar effects for generative benchmarks. We extended our Sec. 2 pretraining sweep into the overtrained regime, pretraining on

$$D(m, N) \stackrel{\text{def}}{=} m \times 20 \times N, \quad (2)$$

tokens per model, where m is the *overtraining multiplier* and N is the number of model parameters. We swept $m \in \{1, 2, 4, 8, 16\}$. Following Sardana et al. (2024); Gadre et al. (2024), we term $m = 1$ “compute-optimal training” and term $m > 1$ “overtraining”. Crucially, in Bordt et al. (2025) and here, as the overtraining multiplier increases, the additional tokens are *new, fresh, non-repeated tokens*; this differs from more practical settings where models might see select documents repeated tens-to-hundreds of times (Hernandez et al., 2022) or the entire corpus repeated for 4+ epochs (Muennighoff et al., 2023; Fang et al., 2025).

We find an interesting interaction between contamination and overtraining (Fig. 3): for models with low contamination, cross entropy on the MATH test set decreases with increasing overtraining, but for models with high contamination, cross entropy on the MATH set increases with overtraining. The cross-over point between test set replicas and overtraining multiplier shifts with model size: the crossover point falls from 32 test set replicas for 34M parameter models to 10 replicas for 63M parameter models to 1 replica for 93M parameter models. Thus, as models become larger, the performance boost from contamination diminishes when overtraining with fresh data. Our interpretation is that while more fresh data is generally useful for improving model performance generally, it dilutes the “dose” of the contaminated data, weakening how the model “responds” (as measured by task performance) (Schaeffer et al., 2025b).

Finding #5: Supervised Fine-Tuning on Training Set Has Opposing Effects, Depending on Pretraining Contamination After pretraining, the first post-training step is oftentimes supervised finetuning (SFT) (Wei et al., 2022b; Ouyang et al., 2022). We turned to assessing what effect, if any, SFT has on contaminated pretrained models. Kocyigit

(2025) recently studied this question and found that SFT on the *train* set improves performance on the *test* set. However, as a key methodological difference, Kocyigit (2025) induced test set contamination via continued pretraining (Jin et al., 2022; Jang et al., 2022; Ibrahim et al., 2024; Parmar et al., 2024; Yildiz et al., 2025), whereas we introduced test set contamination uniformly throughout pretraining.

One might expect that SFTing on the train set should increase test set performance across the board. We discover that, surprisingly, the opposite is sometimes true: SFTing on the train set can both help and hurt model performance on the test set, depending on the amount of contamination in pretraining (Fig. 4). For models with no or low (< 10) test set contamination, SFTing on the train set significantly reduces loss on the test set (purple). At 10 test set replicas, SFT has no effect (aqua), but as contamination increases, SFTing on the train set significantly increases loss on the test set (yellow, green). We conjecture that during SFT, contaminated models learn to generalize but also forget their contaminated pretraining data, and the effects of contaminated test set data are more impactful than generalization for small models, leading to a net increase in test loss. As studied in Section 3, the benefits of contamination on test loss dwarf those of generalization (which would be improved by SFT on the train set). Therefore, the net impact of SFTing on the train set for highly contaminated models is counter-intuitively to increase cross entropy loss.

5. Inference: Temperature & Solution Length

Finally, we arrive at the last stage of the model lifecycle: inference. When models are deployed for generation, inference-time parameters—particularly sampling temperature and the length of generated solutions—introduce new considerations for how contamination manifests.

Finding #6: High Temperature Sampling Mitigates Gains from Contamination We evaluated the pretrained models using temperature-only sampling, sweeping from 0 (“greedy”) to 1.5. We observed that Math Verify scores remain stable between greedy decoding and low-temperature sampling ($\tau \leq 0.56$) (Fig. 5, left and center). However, as temperature increases beyond this point, performance degrades quickly. For example, increasing temperature from 0 to 1 causes performance at high contamination levels (1000 replicas) to collapse by a factor of 40, down to the baseline performance of uncontaminated models. This suggests that contamination-driven performance is brittle; small adjustments in inference settings can eliminate the benefits of contamination almost entirely.

Finding #7: Longer Solution Length Mitigates Gains from Contamination To understand how solution length

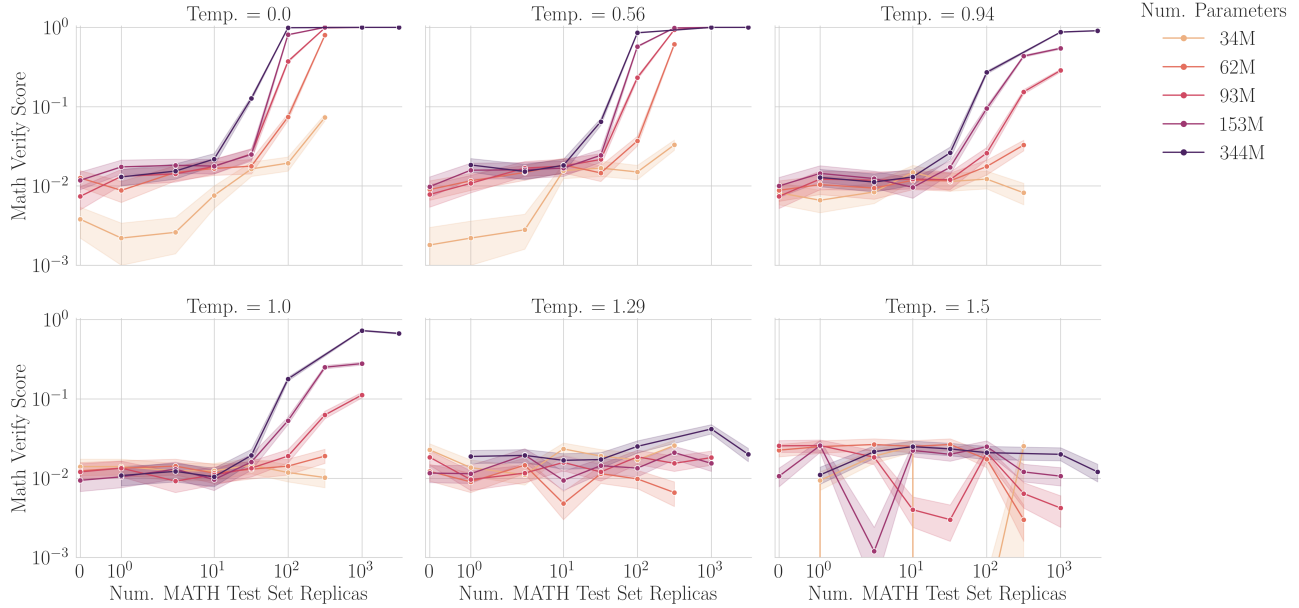


Figure 5. Sampling Temperature Degrades Performance, Particularly for Contaminated Models. We report Math Verify scores as a function of test set replicas and model size across six sampling temperatures. As sampling temperature increases, Math Verify scores drop precipitously, falling from near 100% to under 1% in many configurations. The penalty for high temperature is disproportionately larger for highly contaminated models: increasing temperature from 0 to 1 reduces performance by a factor of ~ 2 at low contamination levels (≤ 10 replicas), whereas it reduces performance by a factor of up to 40 at high contamination levels (1000 replicas).

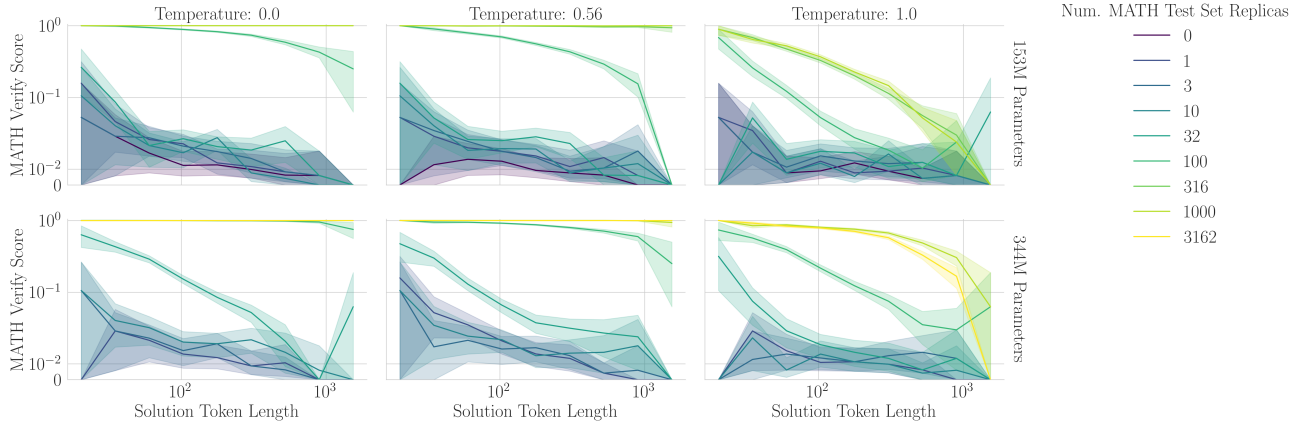


Figure 6. Performance Declines with Increasing Solution Token Length. Math Verify Scores decrease with solution length, with the effects modulated by temperature. At high rates of contamination and higher temperatures, math verify scores fall exponentially with the solution length. At lower temperatures, and high rates of contamination, solution length has almost no effect. Models exposed to less contamination also exhibit declining scores with longer solution lengths, but the dropoff is concave up and occurs much more slowly. Shaded regions represent 95% confidence intervals.

constrains performance, we binned problems into 10 log-spaced intervals ranging from the shortest (15 tokens) to the longest (1949 tokens). We found that Math Verify scores decrease significantly as solution length increases (Fig. 6). We identified a striking shift in the functional form of this decay based on contamination level: At higher contamination levels, the decay follows an approximate power law, but at lower levels of contamination, performance decays exponentially quickly to the noise floor. This suggests that

maintaining a coherent memorized chain becomes increasingly difficult as the sequence grows, and also aligns with findings that longer sequences require more repetitions to be memorized (Jiang et al., 2025; Lu et al., 2024), though we demonstrate this here through controlled pretraining.

Furthermore, solution length interacts with sampling temperature: For short solutions (≤ 100 tokens) at high contamination (316 replicas), raising the temperature from 0 to 1.0



*Figure 7. **Solution Length and Temperature Together Govern How Quickly Memorization Decoheres.** **Top:** Most models and contamination pairs exhibit scaling laws (Eqn. 3) as a function of the sequence length. For exceptions, see Appendix D. **Bottom:** The cumulative probability of generating a memorized solution can exist in one of three regimes: (1) Exponentially Fast Decoherence, (2) Brittle Memorization, and (3) Deterministic Lock-In. Sampling temperature can shift a model between regimes.*

drops accuracy by $\sim 45\%$ on the largest model. However, for solutions of 400 tokens, the same temperature increase causes accuracy to drop by nearly 100%.

Finding #8: Temperature and Solution Length Together Govern the Survival Process Generative evaluations differ fundamentally from discriminative tasks because they require the model to maintain a coherent trajectory over a

sequence of length T . For models that solve the task via memorizing solutions, we can characterize how the model must “survive” the risk of decoherence at every token step t via studying the probabilities under teacher forcing.

To begin, motivated by prior work on in-context scaling laws (????), we find that for most model sizes and number of test set replicas, the negative log likelihood scales with

Figure 8. Three Regimes of Memorization.

the token index as a power law plus an irreducible error:

$$\mathcal{L}_t(N, R) \approx E(N, R) + A(N, R) \cdot t^{-\alpha(N, R)}. \quad (3)$$

However, we do find a small number of notable exceptions (Appendix. D). The probability of successfully generating a solution of length T can then be approximated as:

$$P(T) \approx \exp\left(-E(T-1) - \frac{A}{1-\alpha}(T^{1-\alpha} - 1)\right). \quad (4)$$

To explain intuitively, there are two opposing forces at the intersection of memorization and generation: The probability of generating a sequence decays geometrically (exponentially) with sequence length, making longer sequences harder to generate, but the the probability of knowing the correct next token grows geometrically with sequence length, making predicting the next token easier. These two rates reveal three distinct regimes of memorization:

Regime I: Exponentially Fast Decoherence ($E > 0$). In the absence of meaningful contamination, the irreducible error term E dominates. The survival probability decays exponentially ($P(T) \sim e^{-E \cdot T}$), rendering the generation of long solutions statistically improbable.

Regime II: Brittle Memorization ($E \approx 0, \alpha \leq 1$). At intermediate contamination, the model minimizes the noise floor ($E \rightarrow 0$) but learns the sequence slowly. The probability follows a Weibull-like stretched exponential ($P(T) \sim e^{-kT^{1-\alpha}}$). We characterize this as *brittle memorization*: while the model can accurately generate moderately long sequences, its grasp on the solution is fragile, and the cumulative probability of success eventually vanishes.

Regime III: Deterministic Lock-In ($E \approx 0, \alpha > 1$). At high contamination, we observe a critical phase transition: $\alpha > 1$ flips the sign of the exponent in Eq. 4, causing the penalty term $T^{1-\alpha}$ to decay to zero. Consequently, $P(T)$ converges to a non-zero constant even as $T \rightarrow \infty$. In this regime, the model’s certainty increases faster than the sequence accumulates risk, enabling the *deterministic lock-in* of arbitrarily long solutions.

The Effect of Sampling Temperature. While the native scaling α is a property of the trained model, inference behavior is heavily modulated by the sampling temperature τ . The dependence arises from the mechanics of the softmax distribution: in the regime of memorization, the model’s unnormalized confidence (the logit gap) grows logarithmically with context length ($\Delta z_t \propto \alpha \ln t$). Since temperature acts as a scalar divisor on logits ($z \rightarrow z/\tau$), it modifies the rate

at which probability mass concentrates on the memorized path. This yields an *effective* scaling exponent:

$$\alpha_{\text{eff}}(\tau) \approx \alpha/\tau. \quad (5)$$

This relationship reveals that temperature can artificially shift a model between regimes. Low-temperature sampling ($\tau < 1$) inflates the exponent ($\alpha_{\text{eff}} > \alpha$), potentially pushing a model from **Brittle Memorization** into pseudo-stable **Deterministic Lock-In**. This exposes the fragility of memorization: a model operating in **Deterministic Lock-In** ($\alpha_{\text{eff}} > 1$) at greedy settings ($\tau \rightarrow 0$) can be instantaneously reverted to **Brittle Memorization** ($\alpha_{\text{eff}} < 1$) simply by increasing τ . For example, if a model has a natural scaling $\alpha = 0.8$, greedy decoding effectively amplifies this to $\alpha_{\text{eff}} \gg 1$, feigning robust knowledge. However, sampling at $\tau = 1.0$ exposes the true exponent $\alpha < 1$, causing failure for long sequences. This mechanism explains why high-temperature sampling acts as a “truth serum,” decoupling the gains of contamination from robust generalization.

6. Discussion

Benchmarks serve as our primary proxies for AI capabilities in the wild; test set contamination breaks this proxy, creating a dangerous “illusion of competence.” While prior work has extensively documented this phenomenon in discriminative tasks, this work provides the first comprehensive quantification of contamination mechanics in the generative regime. We conclude that while generative contamination shares the superficial characteristic of inflating scores, the underlying mechanism is distinct: it relies on fragile, verbatim memorization of long token chains that behaves fundamentally differently from robust reasoning.

Our application of neural scaling laws reveals the economic magnitude of this threat. Under standard scaling law assumptions, we demonstrate that a single replica of the test set allows a model to bypass the estimated “irreducible error” barrier, effectively simulating performance that would otherwise require an infinite amount of compute on uncontaminated data. However, this performance is brittle. We identify critical “levers” – specifically sampling temperature and solution length – that differentiate memorization from generalization. Unlike robust reasoning, which survives stochastic sampling, contamination-driven performance collapses under high temperatures. Similarly, the shift from power law to exponential decay in performance as solution length increases signals the limits of the model’s ability to hold a memorized chain without decoherence.

Perhaps most critically for practitioners, standard training pipelines can mask or expose this issue in counter-intuitive ways. The interaction between contamination and further training is governed by a “dose-response” effect (Schaeffer et al., 2025b): overtraining on fresh data dilutes the contam-

ination, and Supervised Fine-Tuning (SFT) on valid training data can actually *degrade* test performance by overwriting memorized test samples. This implies that a drop in test accuracy after SFT, usually a sign of alignment tax or forgetting, may actually be a positive signal of decontamination.

Limitations We focused on a single generative benchmark, MATH, to enable automatic verification and controlled contamination. Consequently, our findings may not fully capture how contamination behaves for other tasks such as coding or creative writing. Additionally, our experiments utilized decoder-only dense transformer models (Qwen 3) up to 344M parameters; results at this scale may not extrapolate to larger models or other architectures. Finally, our pretraining corpus represents a specific mixture of web-crawl data; specialized or heavily filtered corpora could alter the base difficulty of memorization and the specific thresholds at which contamination effects become visible.

Future Directions Our findings suggest several avenues for future research in generative evaluation:

- **Reconciling Memorization Thresholds:** Our discovery that a single test set replica drives loss below the irreducible error contrasts with recent assertions that single-shot verbatim memorization is an “illusion” (Huang et al., 2024) or that membership inference is fundamentally capped (Hayes et al., 2025). Future work should investigate what the explanatory differences are, though recent work suggests memorization depends on frequency relative to corpus size (Schaeffer et al., 2025b; Wei et al., 2025).
- **Inference-Time Detection Methods:** Since high-temperature sampling and long-context generation disproportionately harm contaminated models, future work could develop lightweight “stress tests” that sweep temperature or solution length to detect contamination without access to the pretraining corpus.
- **Architectural Susceptibility:** We studied dense models, but it remains an open question whether Mixture-of-Experts (MoEs) or State Space Models (SSMs) exhibit different memorization capacities.
- **Harder Benchmarks:** We hypothesize that contamination efficacy is inversely correlated with problem compressibility. Investigating whether harder benchmarks (e.g., AIME) require higher “doses” of leakage to achieve the same performance boost would be a valuable contribution to benchmark design.

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A. Related Work

Data Contamination and its Consequences Test set contamination, where benchmark data is included in pretraining corpora, is widely recognized as a threat to valid model evaluation, as it can lead to inflated performance metrics. Numerous survey and position papers have documented the various ways contamination can occur and have called for routine audits and transparent reporting for all benchmarks (Sainz et al., 2023; 2024; Deng et al., 2024a; Xu et al., 2024a; Reuel et al., 2025). Empirical studies of large web-scale datasets have confirmed significant overlap and duplication between training and test sets (Dodge et al., 2021). Research focused on ensuring benchmark integrity has identified multiple ways that language models might “cheat” on evaluations if contamination is not properly managed (Zhou et al., 2023; Dong et al., 2024). For instance, analyses of popular mathematics benchmarks have revealed signals of data leakage and potential overfitting (Zhang et al., 2024a). Ongoing community efforts and open-source audits continue to measure the extent of contamination across different models and datasets (Li et al., 2024). The risks extend beyond evaluation integrity; scaling studies indicate that poisoning risks increase with model size, as larger models learn harmful behaviors from minuscule amounts of poisoned data far more rapidly than smaller models, underscoring the necessity of robust data curation (Bowen et al., 2025). As a cautionary illustration, Schaeffer (2023) demonstrated that pretraining on the test set is a trivial path to strong benchmark performance, reinforcing the importance of rigorous decontamination and auditing.

Controlled Contamination During Pretraining A line of research directly investigates the causal effects of contamination by intentionally adding benchmark data to pretraining corpora and observing the results. Magar & Schwartz (2022) interleaved task-specific datasets into a general text corpus during pretraining, varying the duplication rate of the leaked examples. They differentiated between “memorization” (storing examples) and “exploitation” (using stored examples to boost test scores), finding that both model size and the number of repetitions increased exploitation. Jiang et al. (2024) pretrained models from scratch on corpora containing either only the inputs (“text-only”) or the full input-output pairs (“ground-truth”) of benchmark examples, sweeping the contamination frequency. They observed significant performance gains when ground-truth pairs were used and showed that simple n-gram-based detection methods could be bypassed by paraphrasing or partial data leaks. The problem also transcends language barriers; Yao et al. (2024) demonstrated a cross-lingual contamination channel where continuing to pretrain a model on non-English translations of English benchmarks led to material improvements on the original English tests, a form of contamination that string-matching would not detect. At a larger scale, Bordt et al. (2025) varied the repetition count of leaked examples, model size (up to 1.6B parameters), and the total training token budget, finding that performance scales predictably with size and repetition. They also showed that sufficiently long training on abundant unique data could mitigate or even reverse the effects of earlier contamination. In the context of machine translation, Kocyigit et al. (2025) injected source-target pairs into the pretraining data of 1B and 8B parameter models, quantifying significant overestimation in BLEU scores, with larger models and low-resource languages showing more pronounced effects. Together, these causal intervention studies provide clear evidence that language models memorize and leverage benchmark data when it is present during pretraining.

Repeated Data and Memorization Dynamics Closely related is the study of memorization dynamics, particularly how repeated data affects model behavior. Hernandez et al. (2022) trained models where a small portion of the data was repeated many times, observing strong double descent phenomena (Advani et al., 2020; Belkin et al., 2019; Adlam & Pennington, 2020; Bordelon et al., 2020; Schaeffer et al.) and showing that repeating just 0.1% of tokens 100 times could significantly degrade generalization. Studies tracking exact-sequence memorization have shown that larger models not only memorize more content and at a faster rate but also forget less over the course of training (Tirumala et al., 2022). Carlini et al. (2023) quantified log-linear relationships between verbatim generation and model size, data duplication count, and prompt length. Other work has explored the feasibility of “forecasting” whether a model will memorize a specific string, finding that accurate prediction is possible but may require a substantial portion of the target model’s pretraining compute (Biderman et al., 2023). Beyond explicit repetition, Duan et al. (2025) discovered “latent memorization”, where memorized sequences that are not obvious at a final checkpoint can persist and be revealed later, posing privacy risks. Finally, memorization appears to be task-dependent: Wang et al. (2025) observed stronger memorization for knowledge-intensive QA, whereas machine translation and mathematical reasoning demonstrated greater novelty. Memorization also interacts with logical reasoning; using dynamically generated puzzles, Xie et al. (2025) showed that models could be fine-tuned to perfectly memorize training examples yet failed on slight variations, even as their genuine reasoning abilities also improved, revealing a complex balance between the two.

Detecting and Proving Contamination Another significant area of research focuses on detecting or proving test set contamination in existing models. Oren et al. (2023) and Ni et al. (2025) proposed statistical tests with provable control over false positives by testing if a benchmark’s canonical ordering is statistically privileged over random shuffles. Shi et al. (2024) introduced Min- k %-Prob to determine if a sequence likely appeared in pretraining using only black-box probabilities. Two related works from Golchin & Surdeanu (2023; 2024) frame detection as a multiple-choice “quiz” and use temporal information about model training windows versus benchmark release dates, a strategy also used by Roberts et al. (2024). Broader audits have aimed to quantify leakage and decontamination across a wide range of tasks and models (Xu et al., 2024b; Deng et al., 2024b; Li et al., 2024), while Yang et al. (2023) showed that rephrasing benchmark questions can often bypass n-gram filters. In the domain of code generation, Riddell et al. (2024) quantified contamination in popular coding benchmarks and connected the degree of overlap to performance differences. Matton et al. (2024) cataloged various channels for leakage and released a dataset (LBPP) to help mitigate these issues. Complementing these audits, Yang et al. (2025b) systematically tested fine-grained contamination scenarios in code intelligence across different model types, finding that paired contamination substantially affects LLMs under a pretraining-plus-inference paradigm but has limited effect under a pretrain–finetune–inference pipeline. Other work has also provided instruments for detecting the origins of chain-of-thought sequences (Li et al., 2025).

Preventing Test Set Contamination The growing concern over contamination has spurred the development of new methods for creating benchmarks. These include dynamically updated benchmarks (Jain et al., 2025; Xia et al., 2024; Zhang et al., 2025; Qian et al., 2024) and private or restricted-access benchmarks (Zhang et al., 2024a; Glazer et al., 2025). Recently, Nie et al. (2025) released a benchmark consisting of unsolved scientific questions, which, by its nature, prevents models from being trained on the correct solutions.

Retrieval- and Agent-Time Contamination As model evaluation evolves from static prompting to using tool-augmented agents, the risk of contamination expands. Han et al. (2025) introduced search-time contamination, where an agent retrieves benchmark questions and answers from the web during its evaluation process, which can artificially inflate its performance.

Membership Inference Attacks The field of Membership Inference Attacks (MIA) aims to determine if a specific data point was used to train a model, given only access to the model itself (Shokri et al., 2017). This is highly relevant to contamination, as detection can be viewed as an MIA problem. While the MIA literature is extensive in computer vision (Yeom et al., 2018; Salem et al., 2018; Sablayrolles et al., 2019; Jagielski et al., 2024), it has more recently been applied to language models (Carlini et al., 2021; Zarifzadeh et al., 2023; Shi et al., 2024; Mattern et al., 2023; Li et al., 2023). However, progress in sequence-level MIA for language models has been complicated by issues such as flawed evaluations (Meeus et al., 2024; Zhang et al., 2024b; Jiang et al., 2025). Duan et al. (2024) argue that membership can be inherently “blurry” for natural language. Das et al. (2024) and (Meeus et al., 2024) report that existing MIA testbeds suffer from distribution shifts. Kong et al. (2023) refute MIAs with a theoretical attack, and Liu et al. (2025) and Mangaokar et al. (2025) demonstrate fundamental limitations and exploits of n-gram based methods. Due to these challenges, recent work explores strengthening the membership signal by using multiple correlated sequences as input (Maini et al., 2021; Kandpal et al., 2023; Maini et al., 2024), which aligns more closely with detecting contamination of an entire test set rather than a single example (Golchin & Surdeanu, 2023; Oren et al., 2023).

B. Pretraining Implementation Details

Model Architecture We pretrained Qwen 3 (Yang et al., 2025a) architecture causal language models from random initialization. Table 2 shows the depth (number of layers) and width (hidden size) configurations for each model size. The intermediate size for the feed-forward layers follows Qwen 3’s formula: $256 \cdot \lfloor (255 + \lfloor 8 \cdot \text{hidden_size}/3 \rfloor) / 256 \rfloor$. All models used Flash Attention 2 (Dao, 2023) and bfloat16 precision.

Table 2. Model architecture configurations following Qwen 3 scaling patterns.

Parameters	Num. Layers	Hidden Size
34M	3	96
62M	5	160
93M	6	224
153M	9	320
344M	14	576

Optimizer and Learning Rate We used the AdamW optimizer (Loshchilov & Hutter, 2019) with HuggingFace defaults (i.e., $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, weight decay = 0). We used linear warmup for 250 steps followed by cosine annealing to zero. Following Shuai et al. (2024), we scaled the batch size with the total number of training tokens D as:

$$\text{tokens per optimizer step} = 3.24 \times 10^3 \times D^{0.264}. \quad (6)$$

The learning rate was scaled with the square root of the batch size: $\eta = 10^{-6} \times \sqrt{\text{tokens per optimizer step}}$. Gradients were clipped to a maximum norm of 1.0.

Data Mixing and Contamination For each configuration, we created a pretraining corpus by mixing documents from FineWeb-Edu-Dedup (Penedo et al., 2024) with replicated copies of the MATH test set. The MATH test set was formatted using the template from EleutherAI’s LM Evaluation Harness: “Problem: {problem}\n\nSolution: {solution}”. For a given number of test set replicas R , we: (1) replicated the tokenized MATH test set R times, (2) computed the remaining token budget as $D_{\text{corpus}} = D_{\text{total}} - R \times |\text{MATH test set}|$, (3) sampled documents from FineWeb-Edu-Dedup to fill D_{corpus} tokens, and (4) shuffled the combined dataset. This ensures that total training tokens remain constant across contamination levels, isolating the effect of contamination from the effect of additional data. Each sequence was truncated to a maximum length of 2048 tokens and terminated with an EOS token.

Distributed Training Training used PyTorch’s DistributedDataParallel (DDP) with the NCCL backend. All experiments were logged to Weights & Biases, and trained models were uploaded to HuggingFace Hub.

For more information, please see our [public GitHub repository](#).

C. Math Verify Scoring Bug

During our experiments, we discovered that EleutherAI’s Language Model Evaluation Harness (Gao et al., 2024) contained a bug in its Math Verify scoring implementation that caused systematically incorrect results. This bug affected the `minerva_math` task and related mathematics evaluation tasks.

The Problem The `minerva_math` task’s utility code called `remove_boxed()` on extracted answers *before* passing them to `math_verify.parse()`. The original implementation was:

```
res = verify(parse(doc["answer"]), parse(candidates))
```

where `doc["answer"]` had already been processed to remove the $\text{\LaTeX } \boxed{\ }$ wrapper.

The `math_verify.parse()` function relies on the `\boxed{ }` notation to properly identify and extract mathematical expressions. When the boxed notation is stripped beforehand:

- `parse("\dfrac{9}{7}")` returns empty results (parsing fails)
- `parse("\boxed{\dfrac{9}{7}}")` correctly returns the parsed expression

This caused Math Verify scores to return 0 even when the model’s answer was mathematically correct. As a diagnostic, we evaluated the benchmark’s own gold reference solutions and observed Math Verify scores of approximately 70%—a clear indication that the scoring mechanism itself was flawed rather than the solutions.

The Fix We reported this issue and a fix was merged that passes the full solution text to `math_verify.parse()`, allowing it to use its built-in extraction heuristics:

```
res = verify(gold=parse(doc["solution"]), target=parse(candidates))
```

By using `doc["solution"]` (the complete solution with `\boxed{ }` intact) instead of the pre-processed `doc["answer"]`, the parser can correctly identify and extract the mathematical expressions.

Implications This bug was present in versions of the evaluation harness prior to v0.4.8 (August 2025). Any research reporting MATH benchmark scores using Math Verify from affected versions may have systematically underestimated model performance. We recommend that researchers using these benchmarks verify they are using a corrected version of the evaluation code.

D. Exceptions to In-Context / Sequence Scaling Laws

In Sec. 5, we reported that most model sizes and number of test set replicas exhibit in-context scaling laws (Eq. 3). While the power law plus irreducible error model provides an excellent fit for the majority of experimental conditions, we observe a systematic deviation in specific edge cases. In these instances, the negative log-likelihood (NLL) *increases* with token index t rather than decaying. We identified significant positive slopes in the NLL-vs-token trajectory for two distinct groups of models.

Group I: Capacity Limitations (Uncontaminated Regime) For the smallest uncontaminated model (34M, $R = 0$), the NLL increases monotonically from token 1 to token 645. This behavior is consistent with limited effective context windows in small architectures. Without the aid of memorization ($R = 0$), the 34M parameter model struggles to model long-range dependencies, causing prediction quality to degrade as the necessary context grows beyond its effective capacity. Larger uncontaminated models (62M+) do not exhibit this behavior, showing standard NLL decay.

Group II: Selection Bias (High Contamination Regime) The most pronounced uptick occurs in the largest, most contaminated models (e.g., 344M, $R = 3162$), where NLL is extremely low ($\sim 10^{-3}$) but rises sharply at late token indices. Our analysis suggests this is driven by **selection bias inherent to the dataset structure**.

Because the MATH dataset contains solutions of varying lengths, the set of problems contributing to the NLL at $t = 800$ is a small, non-random subset of the problems at $t = 100$.

- **Survivor Bias:** Only the 100 longest solutions in the test set reach token 800.
- **Differential Memorizability:** We observe a strong correlation between solution length and residual NLL in the high-contamination regime. The uptick indicates that these long surviving sequences are systematically “harder” to memorize than the shorter sequences that drop out earlier.

We verified that these deviations are not artifacts of data corruption. The token-level sample counts are consistent across all runs, with 100% of sequences ($N=5,000$) present at token 0, dropping to 2% ($N=100$) at token 800. This attrition is strictly deterministic, governed by the length distribution of the MATH dataset.

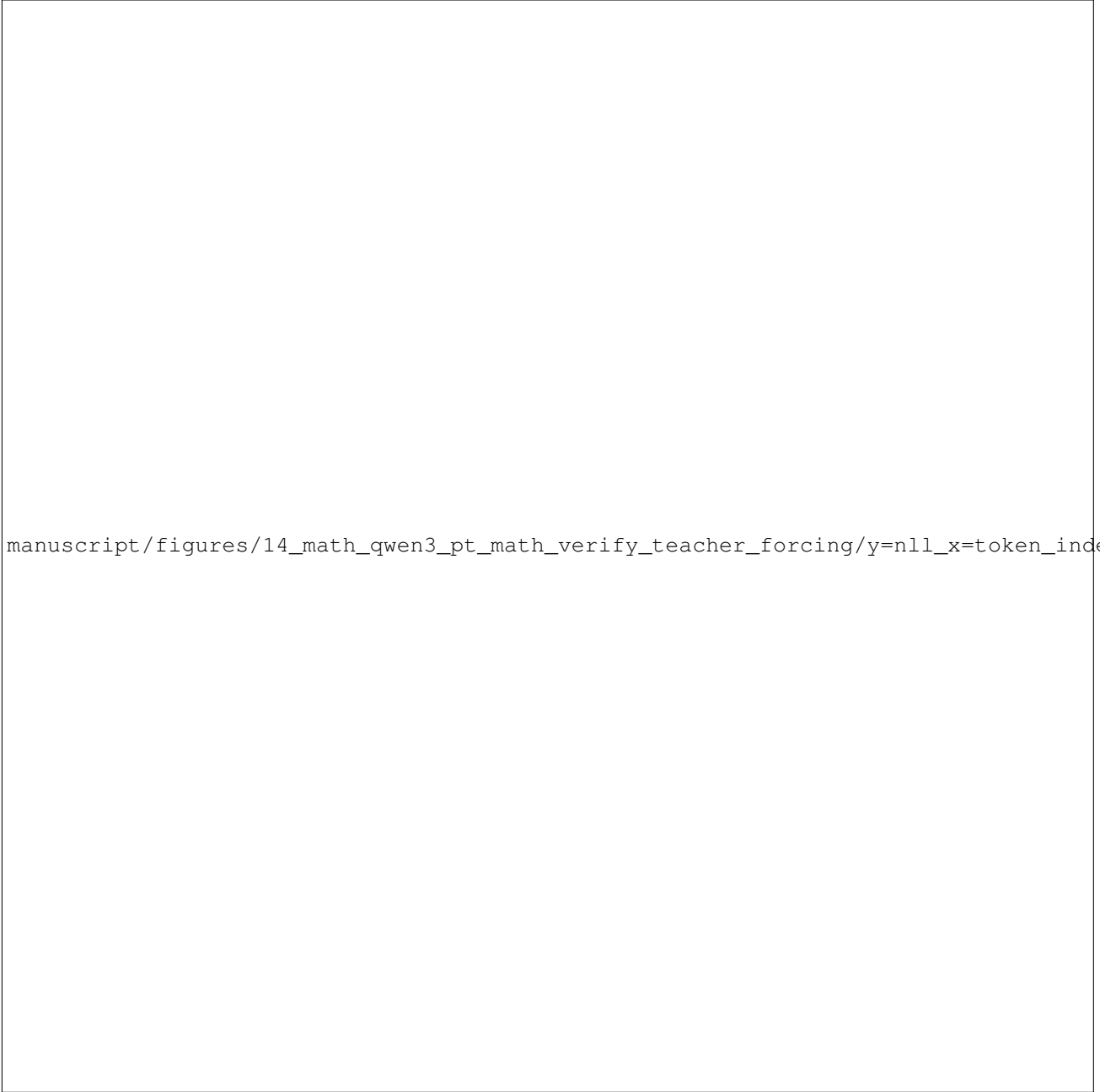


Figure 9. Deviations from In-Context / Sequence Scaling Laws. Two groups of models exhibit increasing negative log likelihoods with increasing token index: smaller uncontaminated models and larger massively contaminated models.



Figure 10. Fit Parameters for In-Context / Sequential Scaling Laws by Model Size and Number of Test Set Replicas.

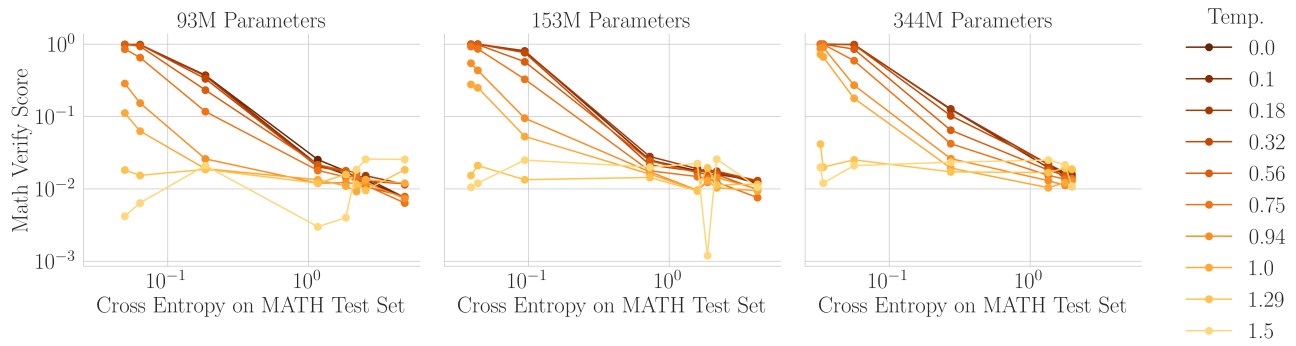


Figure 11. Math Verify Score Is Correlated with Pretraining Loss. Math Verify scores correlate strongly with the cross entropy loss achieved on the MATH Test Set during training, where differences in these graphs are attributable to increased repetition on the benchmark test set. The correlation is significantly weaker for high temperatures and falls to nearly 0 for temperatures above 1.0.